S.M.O.G. – Statistical Modeling of Orbital Gases

Bridging the air quality data gap — using satellites, weather data, and machine learning to predict what sensors currently measure.

GROUP 3

DATA SCIENCE · AIR QUALITY

S.M.O.G Team

1

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Data Scientist and Mechatronics Engineer, expert in sensory data measurement

2

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Data Scientist and Energy Engineer, expert in environmental science

3

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Data Scientist and Biotechnologist, expert in human respiratory system

The Problem: Degrading Air Quality

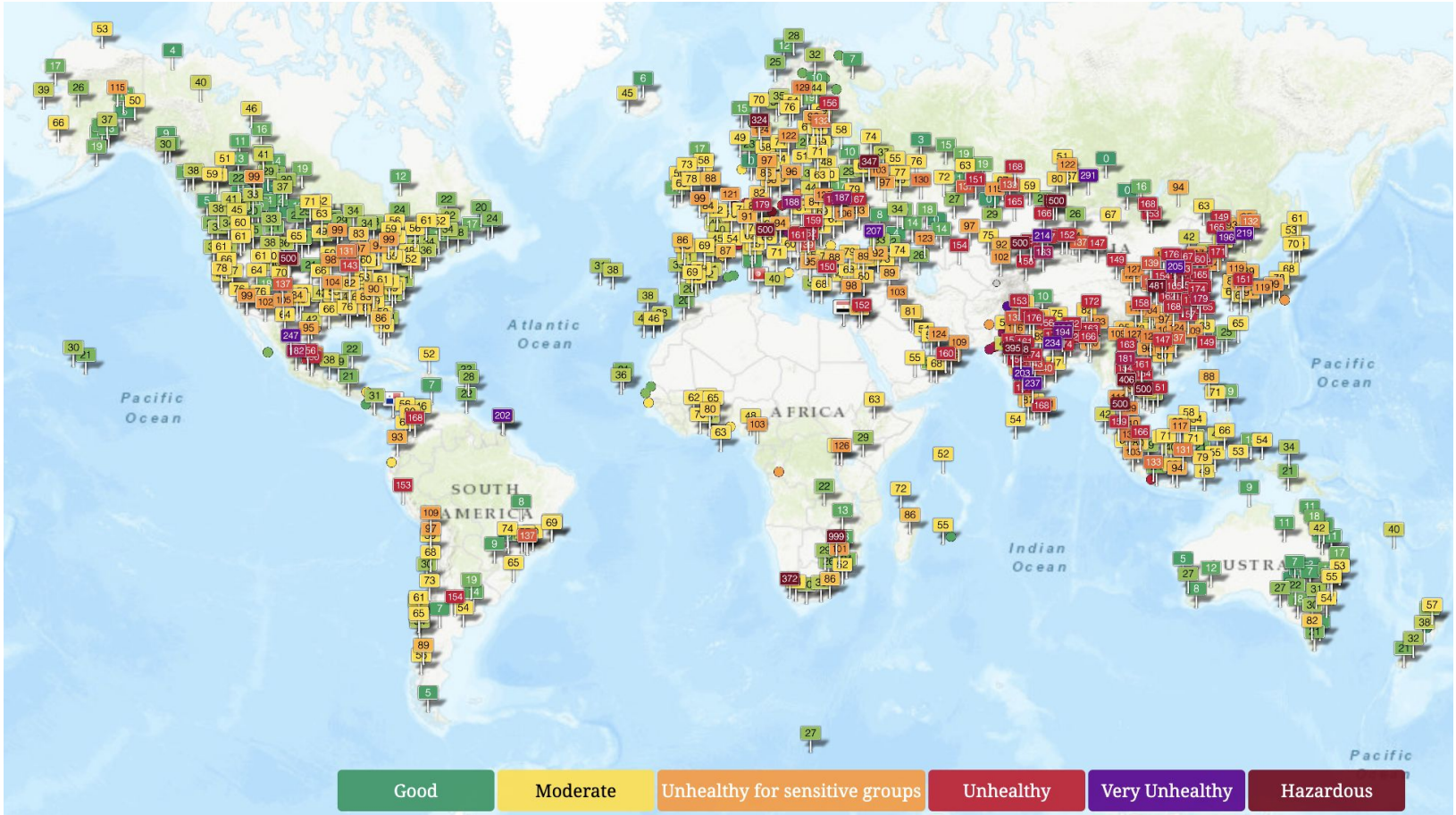
Sensor Scarcity

- Air Quality sensors are expensive
- Vast populations (in Africa) live without this data

PM2.5 Blind Spot

- Without sensors, real-time tracking of PM2.5 is virtually impossible
- PM2.5 is tied directly to respiratory disease

Revealing the geographical Blind Spots



Our Data: A Global View from Above

Sentinel 5P Satellite

- NO_2
- SO_2
- HCHO
- Aerosol
- O3
- CO
- Cloud data
- CH4

Global Weather Data

- Temperature
- Wind speed
- Humidity
- Precipitation

Global City Coverage

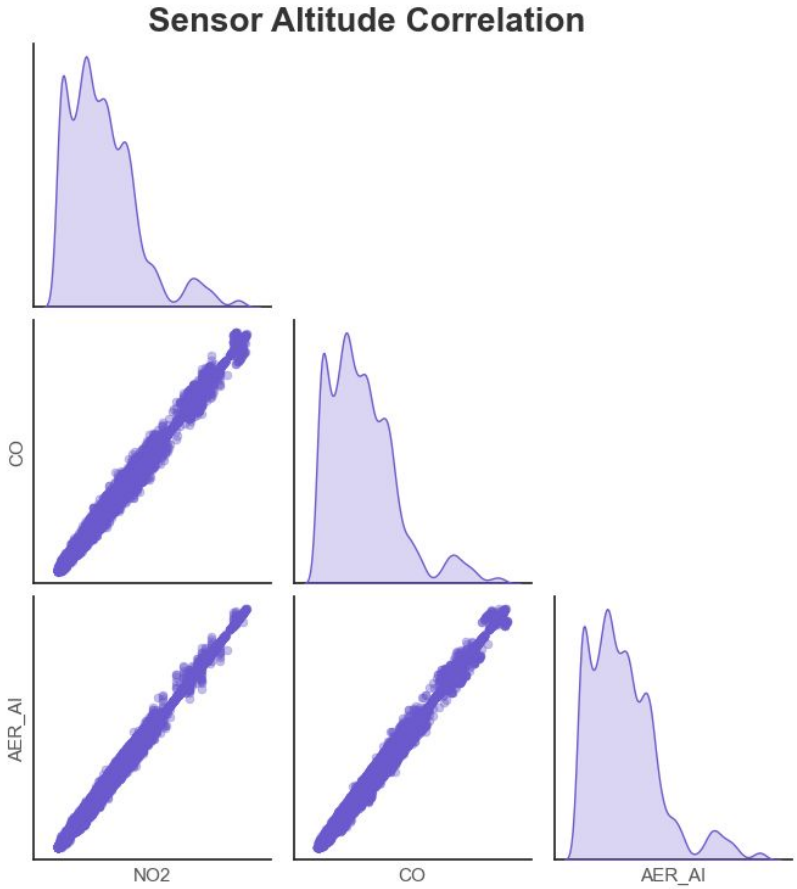
- Hundreds of cities tracked over three months.

Feature Engineering:

1) Addressing multicollinearity

For:

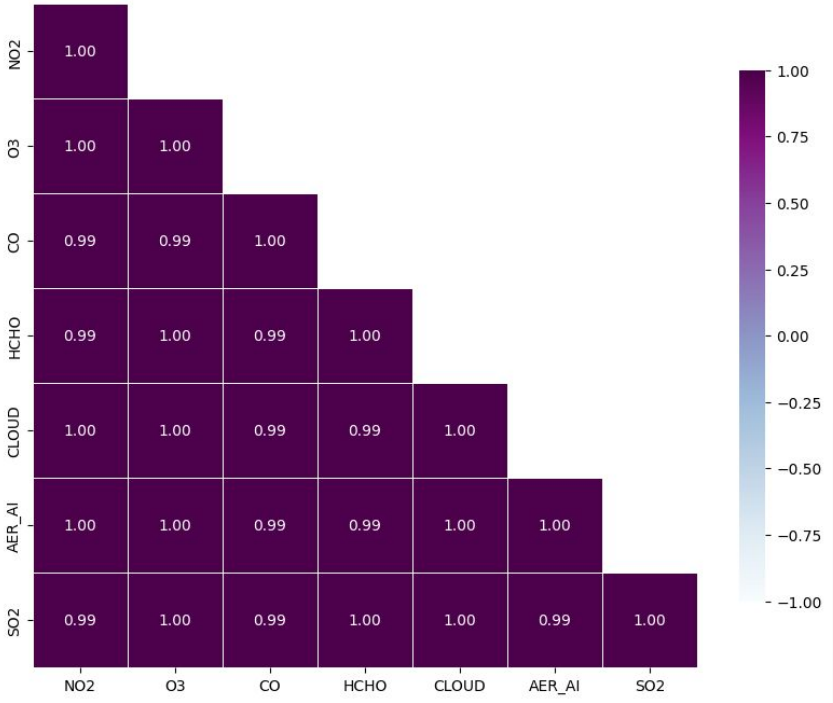
- Sensor altitude
- Cloud coverage
- Sensor & solar angles



Feature Engineering:

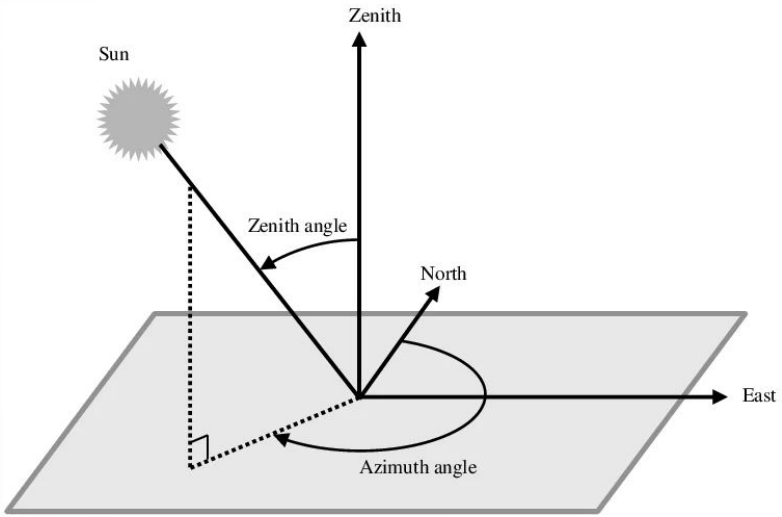
2) Relative Azimuth & Zenith

Solar Zenith Angle Correlation for the Pollutants



$$\Delta\text{Azimuth} = |\phi_{solar} - \phi_{sensor}|$$

$$\Delta\text{Zenith} = |\theta_{solar} - \theta_{sensor}|$$



Feature Engineering:

3) Tropospheric Ratio

- Isolate ground-level pollution from stratospheric interference

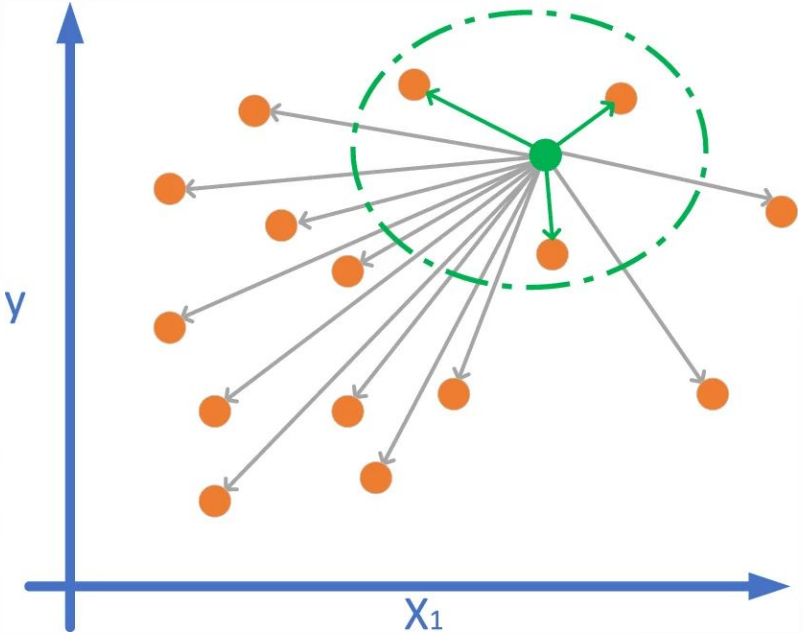
4) Cloud thickness

- Counter potential masking effect of cloud cover.

5) Calculate Air Mass Factors (AMF)

- Correction for the optical path length of light through the atmosphere

The Baseline Model



We used **k-Nearest Neighbors (kNN)** as our baseline model.

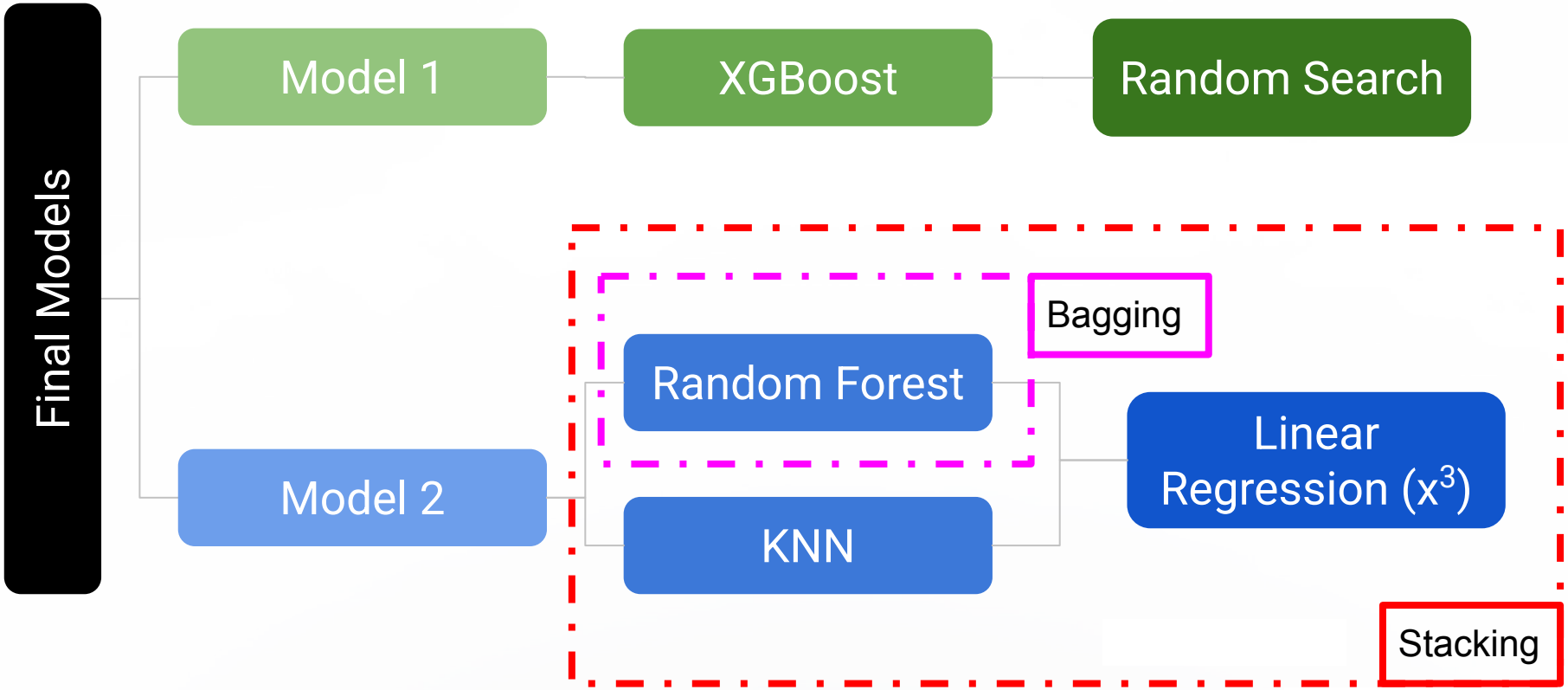
Baseline RMSE :

- ➔ Train: 27.52
- ➔ Validation: 32.79
- ➔ Test: 33.70

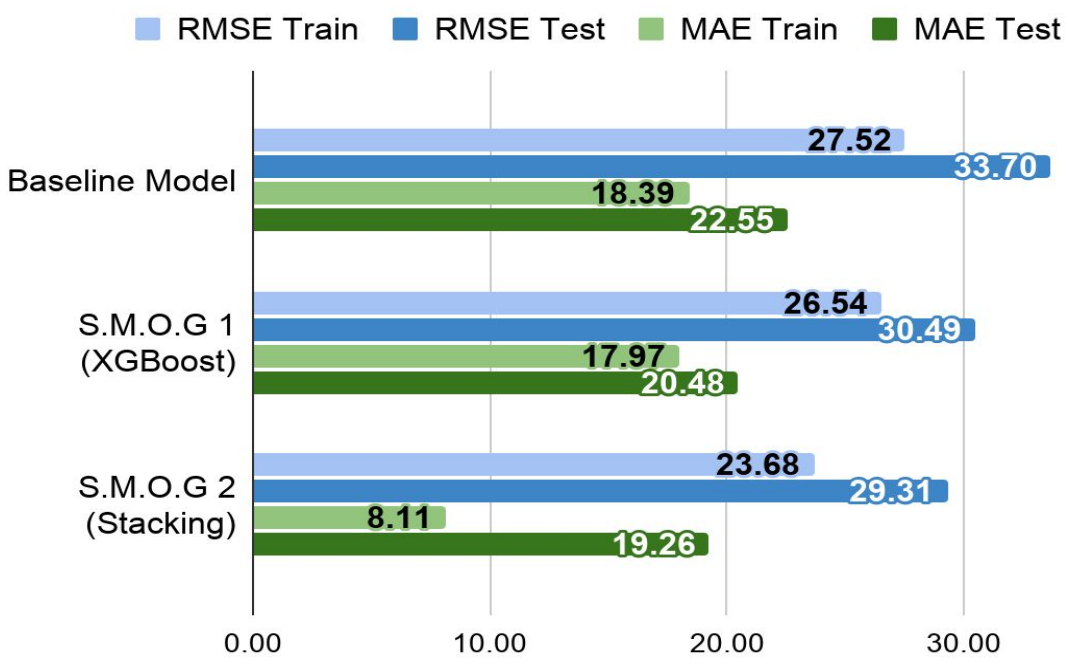
Baseline MAE :

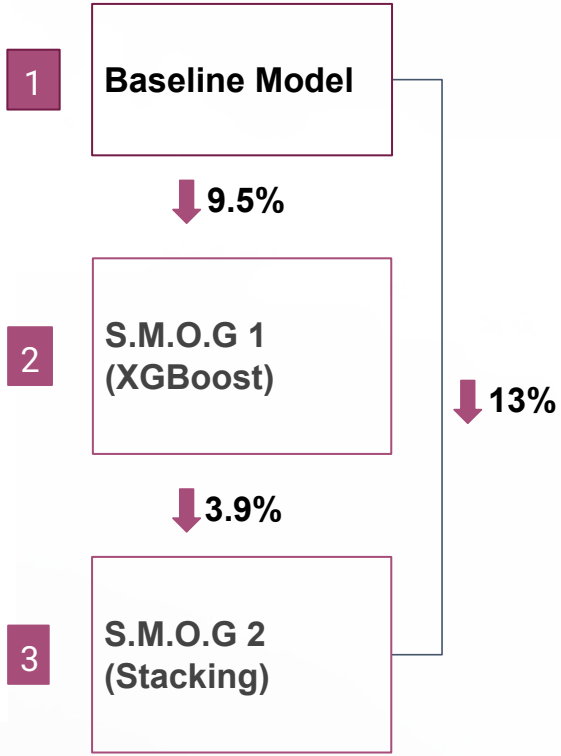
- ➔ Train: 18.32
- ➔ Validation: 22.68
- ➔ Test: 22.55

S.M.O.G Models:


Final Models
Model 1
XGBoost
Random Search
Bagging
Random Forest
Model 2
KNN
Linear Regression (x^3)
Stacking

Model Performance: Baseline vs. S.M.O.G.





Future Horizons: Expanding Our Reach

1. Additional Feature Engineering
2. Utilizing Time Series
3. Using the Max value of Target
4. Testing of Model with more recent data

Democratizing air Quality Information

Addressing the Problem

S.M.O.G. models effectively bridges the sensor gap across Africa

12K+
Monitoring Stations

~13%
Error Reduction
Baseline to S.M.O.G 2

100+
Countries Covered

GROUP 3 — S.M.O.G.

STATISTICAL MODELING OF ORBITAL GASES

Backup Slide

AQI (US)	PM2.5 (μg/m3)	Remark	NAQI	PM2.5 (μg/m3)	Remark
0-50	0-12	Good	0-50	0-30	Good
51-100	12-35	Moderate	51-100	30-60	Satisfactory
101-150	35-55	Unhealthy for Sensitive Individuals	101-200	60-90	Moderate
151-200	55-150	Unhealthy			
201-300	150-250	Very Unhealthy	201-300	90-120	Poor
>300	>250	Hazardous	301-400	120-250	Very Poor
			401-500	>250	Severe